

```
import pandas as pd
```

```
data = pd.read_csv(r"C:\Users\Dhanwantari Devre\Desktop\SQL\dataset_1_202506111712.csv")
```

```
data
```

	destination	passanger	weather	temperature	time \
0	No Urgent Place	Alone	Sunny	55	2PM
1	No Urgent Place	Friend(s)	Sunny	80	10AM
2	No Urgent Place	Friend(s)	Sunny	80	10AM
3	No Urgent Place	Friend(s)	Sunny	80	2PM
4	No Urgent Place	Friend(s)	Sunny	80	2PM
...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM
12680	Work	Alone	Rainy	55	7AM
12681	Work	Alone	Snowy	30	7AM
12682	Work	Alone	Snowy	30	7AM
12683	Work	Alone	Sunny	80	7AM

	coupon expiration	gender	age	maritalStatus	
...	...	...	...	...	
0 \	Restaurant(<20)	1d	Female	21	Unmarried partner
...	...	...	...	...	...
1	Coffee House	2h	Female	21	Unmarried partner
...	...	...	...	...	...
2	Carry out & Take away	2h	Female	21	Unmarried partner
...	...	...	...	...	...
3	Coffee House	2h	Female	21	Unmarried partner
...	...	...	...	...	...
4	Coffee House	1d	Female	21	Unmarried partner
...	...	...	...	...	...
...	...	...	...	...	...
12679	Carry out & Take away	1d	Male	26	Single
...	...	...	...	...	...
12680	Carry out & Take away	1d	Male	26	Single
...	...	...	...	...	...
12681	Coffee House	1d	Male	26	Single
...	...	...	...	...	...
12682	Bar	1d	Male	26	Single
...	...	...	...	...	...
12683	Restaurant(20-50)	2h	Male	26	Single
...	...	...	...	...	...

	CarryAway	RestaurantLessThan20	Restaurant20To50
toCoupon_GEQ5min \			
0	NaN	4~8	1~3
1			
1	NaN	4~8	1~3

1				
2	NaN	4~8	1~3	
1				
3	NaN	4~8	1~3	
1				
4	NaN	4~8	1~3	
1				
...	...	...	...	..
.				
12679	1~3	4~8	1~3	
1				
12680	1~3	4~8	1~3	
1				
12681	1~3	4~8	1~3	
1				
12682	1~3	4~8	1~3	
1				
12683	1~3	4~8	1~3	
1				
toCoupon_GEQ15min toCoupon_GEQ25min direction_same direction_opp				
Y \				
0	0	0	0	1
1				
1	0	0	0	1
0				
2	1	0	0	1
1				
3	1	0	0	1
0				
4	1	0	0	1
0				
...	...	...	...	...
..				
12679	0	0	1	0
1				
12680	0	0	0	1
1				
12681	0	0	1	0
0				
12682	1	1	0	1
0				
12683	0	0	1	0
0				
row_count				
0	1			
1	2			
2	3			

```

3          4
4          5
...      ...
12679     12680
12680     12681
12681     12682
12682     12683
12683     12684

```

```
[12684 rows x 27 columns]
```

```
data[['weather', 'temperature']]
```

```

      weather  temperature
0      Sunny           55
1      Sunny           80
2      Sunny           80
3      Sunny           80
4      Sunny           80
...      ...
12679  Rainy           55
12680  Rainy           55
12681  Snowy           30
12682  Snowy           30
12683  Sunny           80

```

```
[12684 rows x 2 columns]
```

```
data.head(10)
```

```

      destination  passanger  weather  temperature  time \
0  No Urgent Place      Alone  Sunny           55  2PM
1  No Urgent Place  Friend(s)  Sunny           80  10AM
2  No Urgent Place  Friend(s)  Sunny           80  10AM
3  No Urgent Place  Friend(s)  Sunny           80  2PM
4  No Urgent Place  Friend(s)  Sunny           80  2PM
5  No Urgent Place  Friend(s)  Sunny           80  6PM
6  No Urgent Place  Friend(s)  Sunny           55  2PM
7  No Urgent Place    Kid(s)  Sunny           80  10AM
8  No Urgent Place    Kid(s)  Sunny           80  10AM
9  No Urgent Place    Kid(s)  Sunny           80  10AM

```

```

      coupon expiration  gender  age
maritalStatus ... \
0      Restaurant(<20)      1d  Female  21  Unmarried
partner ...
1      Coffee House      2h  Female  21  Unmarried
partner ...
2  Carry out & Take away      2h  Female  21  Unmarried
partner ...

```

3		Coffee House	2h	Female	21	Unmarried
partner	...					
4		Coffee House	1d	Female	21	Unmarried
partner	...					
5		Restaurant(<20)	2h	Female	21	Unmarried
partner	...					
6	Carry out & Take away		1d	Female	21	Unmarried
partner	...					
7		Restaurant(<20)	2h	Female	21	Unmarried
partner	...					
8	Carry out & Take away		2h	Female	21	Unmarried
partner	...					
9		Bar	1d	Female	21	Unmarried
partner	...					

	CarryAway	RestaurantLessThan20	Restaurant20To50	toCoupon_GEQ5min	\
0	NaN	4~8		1~3	1
1	NaN	4~8		1~3	1
2	NaN	4~8		1~3	1
3	NaN	4~8		1~3	1
4	NaN	4~8		1~3	1
5	NaN	4~8		1~3	1
6	NaN	4~8		1~3	1
7	NaN	4~8		1~3	1
8	NaN	4~8		1~3	1
9	NaN	4~8		1~3	1

	toCoupon_GEQ15min	toCoupon_GEQ25min	direction_same	direction_opp	Y
\					
0	0	0	0	1	1
1	0	0	0	1	0
2	1	0	0	1	1
3	1	0	0	1	0
4	1	0	0	1	0
5	1	0	0	1	1
6	1	0	0	1	1
7	1	0	0	1	1
8	1	0	0	1	1
9	1	0	0	1	0

```

row_count
0      1
1      2
2      3
3      4
4      5
5      6
6      7
7      8
8      9
9     10

```

```
[10 rows x 27 columns]
```

```

data['passanger'].unique()
array(['Alone', 'Friend(s)', 'Kid(s)', 'Partner'], dtype=object)
data[data['destination']=='Home']

```

coupon \	destination	passanger	weather	temperature	time
13	Home	Alone	Sunny	55	6PM
Bar					
14	Home	Alone	Sunny	55	6PM
Restaurant(20-50)					
15	Home	Alone	Sunny	80	6PM
Coffee House					
35	Home	Alone	Sunny	55	6PM
Bar					
36	Home	Alone	Sunny	55	6PM
Restaurant(20-50)					
...	...	...	...	...	...
...					
12675	Home	Alone	Snowy	30	10PM
Coffee House					
12676	Home	Alone	Sunny	80	6PM
Restaurant(20-50)					
12677	Home	Partner	Sunny	30	6PM
Restaurant(<20)					
12678	Home	Partner	Sunny	30	10PM
Restaurant(<20)					
12679	Home	Partner	Rainy	55	6PM
Take away					

	expiration	gender	age	maritalStatus	...	CarryAway \
13	1d	Female	21	Unmarried partner	...	NaN
14	1d	Female	21	Unmarried partner	...	NaN
15	2h	Female	21	Unmarried partner	...	NaN
35	1d	Male	21	Single	...	4~8

36	1d	Male	21	Single	...	4~8
...	...	...	...	...	...	...
12675	2h	Male	26	Single	...	1~3
12676	1d	Male	26	Single	...	1~3
12677	1d	Male	26	Single	...	1~3
12678	2h	Male	26	Single	...	1~3
12679	1d	Male	26	Single	...	1~3

	RestaurantLessThan20	Restaurant20To50	toCoupon_GEQ5min	\
13		4~8	1~3	1
14		4~8	1~3	1
15		4~8	1~3	1
35		4~8	less1	1
36		4~8	less1	1
...		...	...	...
12675		4~8	1~3	1
12676		4~8	1~3	1
12677		4~8	1~3	1
12678		4~8	1~3	1
12679		4~8	1~3	1

	toCoupon_GEQ15min	toCoupon_GEQ25min	direction_same	direction_opp
Y \				
13	0	0	1	0
1				
14	1	0	0	1
1				
15	0	0	0	1
0				
35	0	0	1	0
1				
36	1	0	0	1
0				
...	...	...	...	...
...				
12675	1	0	0	1
0				
12676	0	0	1	0
1				
12677	1	1	0	1
1				
12678	1	0	1	0
0				
12679	0	0	1	0
1				

	row_count
13	14
14	15
15	16

```

35      36
36      37
...      ...
12675   12676
12676   12677
12677   12678
12678   12679
12679   12680

```

```
[3237 rows x 27 columns]
```

```
data.sort_values('coupon')
```

coupon \	destination	passanger	weather	temperature	time
11702	Home	Partner	Sunny	30	10PM
Bar					
9930	No Urgent Place	Alone	Snowy	30	2PM
Bar					
10632	Home	Alone	Rainy	55	6PM
Bar					
7997	No Urgent Place	Friend(s)	Rainy	55	10PM
Bar					
11166	Work	Alone	Snowy	30	7AM
Bar					
...	...	...	...	...	...
...					
10476	Home	Alone	Sunny	80	6PM
Restaurant(<20)					
5447	Home	Alone	Sunny	80	10PM
Restaurant(<20)					
10478	Home	Alone	Snowy	30	10PM
Restaurant(<20)					
5440	No Urgent Place	Alone	Sunny	80	2PM
Restaurant(<20)					
0	No Urgent Place	Alone	Sunny	55	2PM
Restaurant(<20)					

	expiration	gender	age	maritalStatus	...	CarryAway \
11702	2h	Female	50plus	Married partner	...	4~8
9930	1d	Female	21	Single	...	gt8
10632	1d	Male	21	Single	...	gt8
7997	2h	Male	26	Unmarried partner	...	4~8
11166	1d	Female	41	Married partner	...	gt8
...	...	...	...	...	...	...
10476	1d	Female	31	Unmarried partner	...	1~3
5447	2h	Female	50plus	Single	...	less1
10478	2h	Female	31	Unmarried partner	...	1~3
5440	2h	Female	50plus	Single	...	less1
0	1d	Female	21	Unmarried partner	...	NaN

	RestaurantLessThan20	Restaurant20To50	toCoupon_GEQ5min	\
11702	1~3	less1	1	
9930	gt8	4~8	1	
10632	less1	less1	1	
7997	never	1~3	1	
11166	1~3	less1	1	
...	...	...	...	
10476	1~3	less1	1	
5447	less1	never	1	
10478	1~3	less1	1	
5440	less1	never	1	
0	4~8	1~3	1	

Y	\	toCoupon_GEQ15min	toCoupon_GEQ25min	direction_same	direction_opp
11702		1	1	0	1
0					
9930		0	0	0	1
0					
10632		1	1	0	1
0					
7997		1	0	0	1
1					
11166		1	1	0	1
0					
...		...	...	...	...
..					
10476		0	0	1	0
1					
5447		0	0	1	0
0					
10478		1	1	0	1
0					
5440		1	0	0	1
0					
0		0	0	0	1
1					

	row_count
11702	11703
9930	9931
10632	10633
7997	7998
11166	11167
...	...
10476	10477
5447	5448
10478	10479
5440	5441

0 1

[12684 rows x 27 columns]

data.rename(columns={'destination':'Destination'})

	Destination	passanger	weather	temperature	time \
0	No Urgent Place	Alone	Sunny	55	2PM
1	No Urgent Place	Friend(s)	Sunny	80	10AM
2	No Urgent Place	Friend(s)	Sunny	80	10AM
3	No Urgent Place	Friend(s)	Sunny	80	2PM
4	No Urgent Place	Friend(s)	Sunny	80	2PM
...	...	...	...	...	...
12679	Home	Partner	Rainy	55	6PM
12680	Work	Alone	Rainy	55	7AM
12681	Work	Alone	Snowy	30	7AM
12682	Work	Alone	Snowy	30	7AM
12683	Work	Alone	Sunny	80	7AM

	coupon expiration	gender	age	maritalStatus	
...	\				
0	Restaurant(<20)	1d	Female	21	Unmarried partner
...					
1	Coffee House	2h	Female	21	Unmarried partner
...					
2	Carry out & Take away	2h	Female	21	Unmarried partner
...					
3	Coffee House	2h	Female	21	Unmarried partner
...					
4	Coffee House	1d	Female	21	Unmarried partner
...					
...	...	...	...	..	...
...					
12679	Carry out & Take away	1d	Male	26	Single
...					
12680	Carry out & Take away	1d	Male	26	Single
...					
12681	Coffee House	1d	Male	26	Single
...					
12682	Bar	1d	Male	26	Single
...					
12683	Restaurant(20-50)	2h	Male	26	Single
...					

	CarryAway	RestaurantLessThan20	Restaurant20To50
toCoupon_GEQ5min \			
0	NaN	4~8	1~3
1			
1	NaN	4~8	1~3
1			

2	NaN	4~8	1~3
1			
3	NaN	4~8	1~3
1			
4	NaN	4~8	1~3
1			
...	...	...	...
.			..
12679	1~3	4~8	1~3
1			
12680	1~3	4~8	1~3
1			
12681	1~3	4~8	1~3
1			
12682	1~3	4~8	1~3
1			
12683	1~3	4~8	1~3
1			
toCoupon_GEQ15min toCoupon_GEQ25min direction_same direction_opp			
Y \			
0	0	0	0
1			
1	0	0	0
0			
2	1	0	0
1			
3	1	0	0
0			
4	1	0	0
0			
...	...	...	...
..			
12679	0	0	1
1			
12680	0	0	0
1			
12681	0	0	1
0			
12682	1	1	0
0			
12683	0	0	1
0			
row_count			
0	1		
1	2		
2	3		
3	4		

```

4          5
...      ...
12679    12680
12680    12681
12681    12682
12682    12683
12683    12684

```

```
[12684 rows x 27 columns]
```

```

sorted(data['occupation'].dropna().unique())
#select occupation from dataset_1 group by occupation

```

```

['Architecture & Engineering',
 'Arts Design Entertainment Sports & Media',
 'Building & Grounds Cleaning & Maintenance',
 'Business & Financial',
 'Community & Social Services',
 'Computer & Mathematical',
 'Construction & Extraction',
 'Education&Training&Library',
 'Farming Fishing & Forestry',
 'Food Preparation & Serving Related',
 'Healthcare Practitioners & Technical',
 'Healthcare Support',
 'Installation Maintenance & Repair',
 'Legal',
 'Life Physical Social Science',
 'Management',
 'Office & Administrative Support',
 'Personal Care & Service',
 'Production Occupations',
 'Protective Service',
 'Retired',
 'Sales & Related',
 'Student',
 'Transportation & Material Moving',
 'Unemployed']

```

```

data.groupby('weather')
['temperature'].mean().to_frame('avg_temp').reset_index()

```

```

  weather  avg_temp
0  Rainy    55.000000
1  Snowy    30.000000
2  Sunny    68.946271

```

```

data.groupby('weather')
['temperature'].count().to_frame('count_temp').reset_index()

```

	weather	count_temp
0	Rainy	1210
1	Snowy	1405
2	Sunny	10069

```
data.groupby('weather')
['temperature'].nunique().to_frame('distinct_count_temp').reset_index()
```

	weather	distinct_count_temp
0	Rainy	1
1	Snowy	1
2	Sunny	3

```
data.groupby('weather')
['temperature'].sum().to_frame('sum_temp').reset_index()
```

	weather	sum_temp
0	Rainy	66550
1	Snowy	42150
2	Sunny	694220

```
data.groupby('weather')
['temperature'].min().to_frame('min_temp').reset_index()
```

	weather	min_temp
0	Rainy	55
1	Snowy	30
2	Sunny	30

```
data.groupby('weather')
['temperature'].max().to_frame('max_temp').reset_index()
```

	weather	max_temp
0	Rainy	55
1	Snowy	30
2	Sunny	80

```
data[data['occupation'] == 'Student'].groupby('occupation').size()
```

```
occupation
Student    1584
dtype: int64
```

```
pd.concat([data, table_to_union])['destination'].drop_duplicates()
```

```
-----
-----
NameError                                Traceback (most recent call
last)
Cell In[68], line 1
----> 1 pd.concat([data, table_to_union])
```

```
['destination'].drop_duplicates()
```

```
NameError: name 'table_to_union' is not defined
```

```
pd.merge(data,table_to_join[['time','part_of_day']],on='time',how='inner')[['destination','time','part_of_day']]
```

```
-----
```

```
-----  
NameError                                Traceback (most recent call  
last)
```

```
Cell In[72], line 1
```

```
----> 1
```

```
pd.merge(data,table_to_join[['time','part_of_day']],on='time',how='inner')[['destination','time','part_of_day']]
```

```
NameError: name 'table_to_join' is not defined
```